Discretization of cosmic-fluid–order-parameter-field model

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1 Presentation of the model

The final spherically symmetric equations in Minkowski space then become

$$-\partial_t^2\phi + \frac{1}{r^2}\partial_r(r^2\partial_r\phi) - \frac{\partial V}{\partial\phi} = \eta\gamma(\partial_t\phi + v\partial_r\phi), \tag{1}$$

$$\partial_t E + \frac{1}{r^2}\partial_r(r^2 E v) + p\big[\partial_t\gamma + \frac{1}{r^2}\partial_r(r^2\gamma v)\big] - \frac{\partial V}{\partial\phi}\gamma(\partial_t\phi + v\partial_r\phi) = \eta\gamma^2(\partial_t\phi + v\partial_r\phi)^2, \tag{2}$$

$$\partial_t Z + \frac{1}{r^2}\partial_r(r^2 Z v) + \partial_r p + \frac{\partial V}{\partial\phi}\partial_r\phi = -\eta\gamma(\partial_t\phi + v\partial_r\phi)\partial_r\phi, \tag{3}$$

where $E \equiv \epsilon\gamma$ and $Z \equiv w\gamma^2 v$.

The equation of state is

$$\epsilon(\phi, T) = 3aT^4 + V(\phi, T) - T\frac{\partial V}{\partial T}, \tag{4}$$

$$p(\phi, T) = aT^4 - V(\phi, T). \tag{5}$$

where $a = (\pi^2/90)g_*$ and $w \equiv \epsilon + p$.

2 Discretization of the equation of motion of the field

In current section we will consider discretizations of the following equation:

$$-\partial_t^2\phi + \frac{1}{r^2}\partial_r(r^2\partial_r\phi) - \frac{\partial V}{\partial\phi} = 0, \tag{6}$$

with Dirichlet boundary conditions:

$$\phi|_{r=0} = \phi_0, \qquad \phi|_{r=\infty} = 0. \tag{7}$$

2.1 Spacial discretization of non-linear wave equation

In order to produce spacial disretizations of eq. (6) one can use Bubnov–Galerkin method. We start with transforming eq. (6) from strong form into weak one by multiplying by generic space-time function $q(t, r)$:

$$\forall_q \int_0^\infty dr r^2 \left[-\partial_t^2 \phi + \frac{1}{r^2} \partial_r (r^2 \partial_r \phi) - \frac{\partial V}{\partial \phi} \right] q = \tag{8}$$

$$\forall_q \int_0^\infty dr r^2 \left[-q \partial_t^2 \phi - \partial_r q \partial_r \phi - q \frac{\partial V}{\partial \phi} \right] + r^2 q \partial_r \phi \Big|_0^\infty = 0 \tag{9}$$

In order to discretize (9) on a finite domain $I = [0, R]$ one need to introduce artificial numerical boundary at $r = R$ with absorbing boundary condition:

According to Bubnov–Galerkin method we search for solution $\phi_d(t, r)$ in subspace of continuous piecewise polynomials of order at most k for variational problem (9) for qs belonging to the same subspace. If integrals in eq. (9) are approximated by the numerical quadrature of the order s, then the order of the method is at least $\min(2k, s)$.

In order to construct base of piecewise polynomials we divide numerical domain I into $N \in \mathbb{N}$ separate periods $I_i := [r_i, r_{i+1}[$ numerated by integer numbers $i \in \{0, \ldots, N-1\}$. On each subset I_i we approximate numerical solution $\phi_d(t, r)$ by a time dependent polynomial $\tilde{\phi}_i^k(t, r)$ with the order at most k. Similarly, we treat qs introducing polynomials $\tilde{q}_i^k(t, r)$. Thus, the weak form of the problem (9) takes the form:

$$\forall_{q_d} \int_0^R dr r^2 \left[-q_d \partial_t^2 \phi_d - \partial_r q_d \partial_r \phi_d - q_d \frac{\partial V}{\partial \phi}(\phi_d) \right] + r^2 q_d \partial_r \phi_d \Big|_0^R = \tag{10}$$

$$\forall_{\tilde{q}_i} \sum_{i=0}^{N-1} \int_{r_i}^{r_{i+1}} dr r^2 \left[-\tilde{q}_i^k \partial_t^2 \tilde{\phi}_i^k - \partial_r \tilde{q}_i^k \partial_r \tilde{\phi}_i^k - \tilde{q}_i^k \frac{\partial V}{\partial \phi}(\tilde{\phi}_i^k) \right] + r^2 q_d \partial_r \phi_d \Big|_0^R = 0 \tag{11}$$

2.1.1 Second order discretization

In order to obtain second order discretization we have to use linear piecewise base functions. We will use Lagrange polynomials:

$$\phi_d(t, r)|_{I_i} = \tilde{\phi}_i^1(t, r) = \phi_i(t) \frac{r_{i+1} - r}{r_{i+1} - r_i} + \phi_{i+1}(t) \frac{r - r_i}{r_{i+1} - r_i}, \tag{12}$$

$$q_d(t, r)|_{I_i} = \tilde{q}_i^1(t, r) = q_i(t) \frac{r_{i+1} - r}{r_{i+1} - r_i} + q_{i+1}(t) \frac{r - r_i}{r_{i+1} - r_i}. \tag{13}$$

For purpose of approximating integrals we will use trapezoid rule:

$$\int_a^b f(x) dx = (b - a) \frac{f(b) + f(a)}{2} + \mathcal{O}\left((b - a)^3\right). \tag{14}$$

Substituting linear piecewise approximations and using trapezoid rule one

gets:

$$\begin{aligned}
&\forall_{q_i} \sum_{i=0}^{N-1} \frac{r_{i+1}-r_i}{2} \left\{ r_i^2 \left[-q_i \partial_t^2 \phi_i - \frac{q_{i+1}-q_i}{r_{i+1}-r_i} \frac{\phi_{i+1}-\phi_i}{r_{i+1}-r_i} - q_i \frac{\partial V}{\partial \phi}(\phi_i) \right] \right. \\
&\qquad \left. + r_{i+1}^2 \left[-q_{i+1} \partial_t^2 \phi_{i+1} - \frac{q_{i+1}-q_i}{r_{i+1}-r_i} \frac{\phi_{i+1}-\phi_i}{r_{i+1}-r_i} - q_{i+1} \frac{\partial V}{\partial \phi}(\phi_{i+1}) \right] \right\} \\
&\qquad + r_N^2 q_N \frac{\phi_N - \phi_{N-1}}{r_N - r_{N-1}} - r_0^2 q_0 \frac{\phi_1 - \phi_0}{r_1 - r_0} = \\
&\forall_{q_i} q_0 \frac{r_1 - r_0}{2} r_0^2 \left[-\partial_t^2 \phi_0 - \frac{\partial V}{\partial \phi}(\phi_0) \right] + q_N \frac{r_N - r_{N-1}}{2} r_N^2 \left[-\partial_t^2 \phi_N - \frac{\partial V}{\partial \phi}(\phi_N) \right] \\
&+ \sum_{i=1}^{N-2} q_i \left[r_i^2 \frac{r_{i+1} - r_{i-1}}{2} \left(-\partial_t^2 \phi_i - \frac{\partial V}{\partial \phi}(\phi_i) \right) + \left(\frac{r_{i+1}^2 + r_i^2}{2} \frac{\phi_{i+1} - \phi_i}{r_{i+1} - r_i} - \frac{r_i^2 + r_{i-1}^2}{2} \frac{\phi_i - \phi_{i-1}}{r_i - r_{i-1}} \right) \right] = 0
\end{aligned} \tag{15}$$

For $i = 1, \ldots, N-2$ we get following spacial discretization:

$$r_i^2 \frac{r_{i+1} - r_{i-1}}{2} \left(-\partial_t^2 \phi_i - \frac{\partial V}{\partial \phi}(\phi_i) \right) + \left(\frac{r_{i+1}^2 + r_i^2}{2} \frac{\phi_{i+1} - \phi_i}{r_{i+1} - r_i} - \frac{r_i^2 + r_{i-1}^2}{2} \frac{\phi_i - \phi_{i-1}}{r_i - r_{i-1}} \right) = 0. \tag{16}$$

Verification: Assuming $r_{i+1} = r_i + \Delta r$ we have:

$$\begin{aligned}
r_i^2 \frac{2\Delta r}{2} \left(-\partial_t^2 \phi_i - \frac{\partial V}{\partial \phi}(\phi_i) \right) + \left(\frac{(r_i + \Delta r)^2 + r_i^2}{2\Delta r} (\phi_{i+1} - \phi_i) - \frac{(r_i - \Delta r)^2 + r_i^2}{2\Delta r} (\phi_i - \phi_{i-1}) \right) &= 0 \\
r_i^2 \Delta r \left(-\partial_t^2 \phi_i - \frac{\partial V}{\partial \phi}(\phi_i) \right) + \left(\frac{2(r_i^2 + r_i \Delta r)}{2\Delta r} (\phi_{i+1} - \phi_i) - \frac{2(r_i^2 - r_i \Delta r)}{2\Delta r} (\phi_i - \phi_{i-1}) \right) &= 0 \\
r_i^2 \Delta r \left(-\partial_t^2 \phi_i - \frac{\partial V}{\partial \phi}(\phi_i) \right) + \left(\frac{2r_i^2(\phi_{i+1} - 2\phi_i + \phi_{i-1}) + 2r_i \Delta r (\phi_{i+1} - \phi_{i-1})}{2\Delta r} \right) &= 0 \\
r_i^2 \Delta r \left(-\partial_t^2 \phi_i - \frac{\partial V}{\partial \phi}(\phi_i) \right) + \frac{r_i^2 (\Delta r)^2 \partial_r^2 \phi_i + 2 r_i (\Delta r)^2 \partial_r \phi_i}{\Delta r} &= 0 \\
- \partial_t^2 \phi_i - \frac{\partial V}{\partial \phi}(\phi_i) + \partial_r^2 \phi_i + \frac{2}{r_i} \partial_r \phi_i &= 0 \\
- \partial_t^2 \phi_i + \frac{1}{r_i^2} \partial_r (r_i^2 \partial_r \phi_i) - \frac{\partial V}{\partial \phi}(\phi_i) &= 0
\end{aligned} \tag{17}$$

2.2 Temporal discretization of non-linear wave equation

The problem that we want to solve can be schematically represented by the following equation:

$$\partial_t^2 \phi_i = -\frac{\partial V}{\partial \phi}(\phi_i) - \Delta_d \phi_i, \tag{18}$$

where $\Delta_d \phi_i$ is a discretization of Laplace operator derived in section 2.1. If we introduce the field velocity $\dot{\phi} := \partial_t \phi$, the spatially discretized non-liner wave equation can be rewritten as set of two first order equations:

$$\begin{aligned}
\partial_t \dot{\phi}_i &= -\frac{\partial V}{\partial \phi}(\phi_i) - \Delta_d \phi_i, \\
\partial_t \phi_i &= \dot{\phi}_i.
\end{aligned} \tag{19}$$

Similarly as in the case of spacial discretizations we device time period of integration $J = [0, T]$ into finite separate subsets $J_j :=]t_j, t_{j+1}[: j \in 0, \ldots, M-1$.

2.2.1 Second order discretization

Second order accurate temporal discretization of non-linear wave equation can be obtained using standard Strömer-Verlet method. It can be formulated in velocity form:

$$\dot{\phi}_{i,j+1/2} = \dot{\phi}_{i,j} - \frac{t_{j+1} - t_j}{2} \left[\frac{\partial V}{\partial \phi}(\phi_{i,j}) - \Delta_d \phi_{i,j} \right], \tag{20}$$

$$\phi_{i,j+1} = \phi_{i,j} + (t_{j+1} - t_j)\dot{\phi}_{i,j+1/2}, \tag{21}$$

$$\dot{\phi}_{i,j+1} = \dot{\phi}_{i,j+1/2} - \frac{t_{j+1} - t_j}{2} \left[\frac{\partial V}{\partial \phi}(\phi_{i,j+1}) - \Delta_d \phi_{i,j+1} \right], \tag{22}$$

or in position form:

$$\phi_{i,j+1/2} = \phi_{i,j} + \frac{t_{j+1} - t_j}{2} \dot{\phi}_{i,j}, \tag{23}$$

$$\dot{\phi}_{i,j+1} = \dot{\phi}_{i,j} - (t_{j+1} - t_j) \left[\frac{\partial V}{\partial \phi}(\phi_{i,j+1/2}) - \Delta_d \phi_{i,j+1/2} \right], \tag{24}$$

$$\phi_{i,j+1/2} = \phi_{i,j+1/2} + \frac{t_{j+1} - t_j}{2} \dot{\phi}_{i,j+1}. \tag{25}$$

$$\tag{26}$$

Velocity Strömer-Verlet can be derived as a variational integrator using continuous Galerkin method. On the other hand derivation of position form requires discontinuous base of functions.

Introducing generic functions of time s_i and u_i we can turn equations (19) into its weak Galerkin form:

$$\forall_{s_i} 0 = \int_0^T dt \left[-\partial_t \dot{\phi}_i - \frac{\partial V}{\partial \phi}(\phi_i) - \Delta_d \phi_i \right] s_i \tag{27}$$

$$\forall_{u_i} 0 = \int_0^T dt \left[\partial_t \phi_i - \dot{\phi}_i \right] u_i. \tag{28}$$

Let's start with velocity form derivation. We use piecewise continues linear functions for $\phi_i(t)$:

$$\phi_i(t)|_{J_j} = \tilde{\phi}_{ij}^{\,1}(t) = \phi_{i,j} \frac{t_{j+1} - t}{t_{j+1} - t_j} + \phi_{i,j+1} \frac{t - t_j}{t_{j+1} - t_j}, \tag{29}$$

and piecewise constant functions for $\dot{\phi}_i$:

$$\dot{\phi}_i(t)\Big|_{J_j} = \tilde{\dot{\phi}}_{ij}^{\,0}(t) = \dot{\phi}_{i,j+1/2} \qquad \dot{\phi}_i(t_j) = \dot{\phi}_{i,j} \qquad \dot{\phi}_i(t_{j+1}) = \dot{\phi}_{i,j+1}. \tag{30}$$

Since with this choice of functions we have to differentiate discontinuous function in the equation (27) we will understand time derivative in its weak sense:

$$\forall_f \int_{x_-}^{x_+} dx f(x)(\partial_x g)(x) = -\int_{x_-}^{x_+} dx (\partial_x f)(x) g(x) + f(x_+)g(x_+) - f(x_-)g(x_-). \tag{31}$$

Using probe functions in the form:

$$s_i(t)|_{J_j} = \tilde{s}_{ij}^1(t) = s_{i,j}\frac{t_{j+1}-t}{t_{j+1}-t_j} + s_{i,j+1}\frac{t-t_j}{t_{j+1}-t_j}, \quad u_i(t)|_{J_j} = u_{i,j+1/2}, \quad (32)$$

and trapezoid rule to approximate the integrals we get:

$$\forall_{s_i} 0 = \sum_{j\in 0,M-1} (s_{i,j+1} - s_{i,j})\dot{\phi}_{i,j+1/2} \quad (33)$$

$$- \frac{t_{j+1}-t_j}{2}\left[s_{i,j}\left(\frac{\partial V}{\partial \phi}(\phi_{i,j}) - \Delta_d \phi_{i,j}\right)\right. \quad (34)$$

$$\left. + s_{i,j+1}\left(\frac{\partial V}{\partial \phi}(\phi_{i,j+1}) - \Delta_d \phi_{i,j+1}\right)\right] \quad (35)$$

$$- s_{i,j+1}\dot{\phi}_{i,j+1} + s_{i,j}\dot{\phi}_{i,j}, \quad (36)$$

$$\forall_{u_i} 0 = \sum_{j\in 0,M-1} u_{i,j+1/2}\left[\frac{\phi_{i,j+1}-\phi_{i,j}}{t_{j+1}-t_j} - \dot{\phi}_{i,j+1/2}\right]. \quad (37)$$

Its easy to verify that these equations leads to velocity form of Strömer-Verlet method.

Similar results can be obtained for position Strömer-Verlet method choosing piecewise constant functions for ϕ and piecewise linear functions for $\dot{\phi}$. In this case time derivative in the equation (28) requires interpretation in the weak sense.

References